Passing to the limit we get

$$K_1(\gamma_0) \le K_1(\widetilde{\gamma}) = \mathscr{T}_c(\mu,\nu),$$

which implies that γ_0 is also optimal.

Third step: $\gamma_0 = \overline{\gamma}$. Take any optimal transport plan γ (for the cost K_1) from μ to ν. Set $\gamma^n = (\mathrm{id}\times p_n)_\#\gamma$, where $(\mathrm{id}\times p_n)(x,y) := (x, p_n(y))$. We have $(\pi_y)_\#\gamma^n = \nu^n$. Then we have

$$\mathscr{T}_c(\nu_\varepsilon,\nu) + \varepsilon K_1(\gamma_\varepsilon) + \varepsilon^2 K_2(\gamma_\varepsilon) \le \mathscr{T}_c(\nu^n,\nu) + \varepsilon K_1(\gamma^n) + \varepsilon^2 K_2(\gamma^n) + C\varepsilon^{3d+3} n^d.$$

Moreover, using the triangle inequality of Corollary 3.2, we infer

$$K_1(\gamma_\varepsilon) \ge \mathscr{T}_c(\mu,\nu_\varepsilon) \ge \mathscr{T}_c(\mu,\nu) - \mathscr{T}_c(\nu_\varepsilon,\nu).$$

We also have

$$K_1(\gamma^n) \le K_1(\gamma) + \int |p_n(y) - y|\,\mathrm{d}\gamma(x,y) \le K_1(\gamma) + \frac{1}{n} = \mathscr{T}_c(\mu,\nu) + \frac{1}{n}.$$

Hence we have

$$(1-\varepsilon)\mathscr{T}_c(\nu_\varepsilon,\nu) + \varepsilon\mathscr{T}_c(\mu,\nu) + \varepsilon^2 K_2(\gamma_\varepsilon) \le \frac{1}{n} + \varepsilon\mathscr{T}_c(\mu,\nu) + \frac{\varepsilon}{n} + \varepsilon^2 K_2(\gamma^n) + C\varepsilon^{3d+3} n^d.$$

Getting rid of the first term (which is positive) in the left hand side, simplifying $\varepsilon\mathscr{T}_c(\mu,\nu)$, and dividing by ε^2, we get

$$K_2(\gamma_\varepsilon) \le \frac{1+\varepsilon}{n\varepsilon^2} + K_2(\gamma^n) + C\varepsilon^{3d+1} n^d.$$

Here it is sufficient to take $n \approx \varepsilon^{-3}$ and pass to the limit to get

$$K_2(\gamma_0) \le K_2(\gamma),$$

which is the condition characterizing $\overline{\gamma}$ (K_2-optimality among plans which are K_1-minimizers). □ □

This approximation result will be useful in Chapter 4. It is mainly based on the fact (already mentioned in Section 3.1.2) that, when we minimize the transport cost $|x-y| + \varepsilon|x-y|^2$, we converge to the solution of the secondary variational problem, i.e. to the ray-monotone transport map. As we said, any kind of strictly convex perturbation should do the same job as the quadratic one.

Yet, there are other approximations that are as natural as this one, but are still open questions. We list two of them.

Open Problem (stability of the monotone transport): take γ_n to be the ray-monotone optimal transport plan for the cost $|x-y|$ between μ_n and ν_n. Suppose $\mu_n \rightharpoonup \mu, \nu_n \rightharpoonup \nu$ and $\gamma_n \rightharpoonup \gamma$. Is γ the ray-monotone optimal transport plan between μ and ν?

Open Problem (limit of $\sqrt{\varepsilon^2+|x-y|^2}$): take γ_ε to be the optimal transport plan for the cost $\sqrt{\varepsilon^2+|x-y|^2}$ between two given measures μ and ν. If $\mathrm{spt}(\mu)\cap\mathrm{spt}(\nu) =$

$\emptyset$, then γ_ε can be easily proven to converge to the ray-monotone optimal transport plan (because, at the first non-vanishing order in ε, the perturbation of the cost is of the form $|x-y|+\varepsilon^2/|x-y|+o(\varepsilon^2)$, and the function $h(t)=1/t$ is strictly convex on $\{t>0\}$). Is the same true if the measures have intersecting (or identical) supports as well?

This last approximation is very useful when proving (or trying to prove) regularity results (see [211]).

3.2 The supremal case, L^∞

We consider now a different problem: instead of minimizing

$$K_p(\gamma)=\int |x-y|^p \,\mathrm{d}\gamma,$$

we want to minimize the maximal displacement, i.e. its L^∞ norm. This problem has been first adressed in [123], but the proof that we present here is essentially taken from [198].

Let us define

$$\begin{aligned} K_\infty(\gamma) := ||x-y||_{L^\infty(\gamma)} &= \inf\{m\in\mathbb{R} : |x-y|\le m \text{ for } \gamma-a.e.(x,y)\} \\ &= \max\{|x-y| : (x,y)\in \operatorname{spt}(\gamma)\} \end{aligned}$$

(where the last equality, between an L^∞ norm and a maximum on the support, is justified by the continuity of the function $|x-y|$).

Lemma 3.21. *For every $\gamma\in\mathscr{P}(\Omega\times\Omega)$ the quantities $K_p(\gamma)^{\frac{1}{p}}$ increasingly converge to $K_\infty(\gamma)$ as $p\to+\infty$. In particular $K_\infty(\gamma)=\sup_{p\ge1}K_p(\gamma)^{\frac{1}{p}}$ and K_∞ is l.s.c. for the weak convergence in $\mathscr{P}(\Omega\times\Omega)$. Thus, it admits a minimizer over $\Pi(\mu,\nu)$, which is compact.*

Proof. It is well known that, on any finite measure space, the L^p norms converge to the L^∞ norm as $p\to\infty$, and we will not reprove it here. This may be applied to the function $(x,y)\mapsto|x-y|$ on $\Omega\times\Omega$, endowed with the measure γ, thus getting $K_p(\gamma)^{\frac{1}{p}}\to K_\infty(\gamma)$. It is important that this convergence is monotone here, and this is true when the measure has unit mass. In such a case, we have for $p<q$, using Hölder (or Jensen) inequality

$$\int |f|^p\,\mathrm{d}\gamma \le \left(\int |f|^q\,\mathrm{d}\gamma\right)^{p/q}\left(\int 1\,\mathrm{d}\gamma\right)^{1-p/q} = \left(\int |f|^q\,\mathrm{d}\gamma\right)^{p/q},$$

for every $f\in L^q(\gamma)$. This implies, by taking the p-th root, $||f||_{L^p}\le||f||_{L^q}$. Applied to $f(x,y)=|x-y|$ this gives the desired monotonicity.

From this fact, we infer that K_∞ is the supremum of a family of functionals which are continuous for the weak convergence (since K_p is the integral of a bounded continuous function, Ω being compact, and taking the p-th root does not break continuity). As a supremum of continuous functionals, it is l.s.c. and the conclusion follows. □ □

The goal now is to analyze the solution of

$$\min\{K_\infty(\gamma)\ :\ \gamma\in\Pi(\mu,\nu)\}$$

ant to prove that there is at least a minimizer γ induced by a transport map. This map would solve

$$\min\{||\mathrm{T}-\mathrm{id}||_{L^\infty(\mu)}\ ,\ \mathrm{T}_\#\mu=\nu\}.$$

Here as well there will be no uniqueness (it is almost always the case when we minimize an L^∞ criterion), hence we define $O_\infty(\mu,\nu)=\mathrm{argmin}_{\gamma\in\Pi(\mu,\nu)}K_\infty(\gamma)$, the set of optimal transport plans for this L^∞ cost. Note that $O_\infty(\mu,\nu)$ is compact, since K_∞ is l.s.c. (as for $O(\mu,\nu)$). Set now $L:=\min\{K_\infty(\gamma)\ :\ \gamma\in\Pi(\mu,\nu)\}$: we can write

$$\gamma\in O_\infty(\mu,\nu)\Leftrightarrow \mathrm{spt}(\gamma)\subset\{(x,y):|x-y|\le L\}$$

(since any transport plan γ concentrated on the pairs where $|x-y|\le L$ satisfies $K_\infty(\gamma)\le L$ and is hence optimal). We will suppose $L>0$ otherwise this means that it is possible to obtain ν from μ with no displacement, i.e. $\mu=\nu$ and the optimal displacement is the identity.

Consequently, exactly as for the L^1 case, we can define a secondary variational problem:

$$\min\{K_2(\gamma)\,:\,\gamma\in O_\infty(\mu,\nu)\}.$$

This problem has a solution $\overline{\gamma}$ since K_2 is continuous for the weak convergence and $O_\infty(\mu,\nu)$ is compact. We do not know for the moment whether we have uniqueness for this minimizer. Again, it is possible to say that $\overline{\gamma}$ also solves

$$\min\left\{\int_{\Omega\times\Omega}c\,\mathrm{d}\gamma\,:\,\gamma\in\Pi(\mu,\nu)\right\},\qquad\text{where } c(x,y)=\begin{cases}|x-y|^2 & \text{if } |x-y|\le L\\ +\infty & \text{otherwise.}\end{cases}$$

The arguments are the same as in the L^1 case. Moreover, also the form of the cost c is similar, and this cost is l.s.c. as well. Hence, $\overline{\gamma}$ is concentrated on a set $\Gamma\subset\Omega\times\Omega$ which is c-CM. This means

$$(x_1,y_1),(x_2,y_2)\in\Gamma,|x_1-y_2|,|x_2-y_1|\le L\Rightarrow(x_1-x_2)\cdot(y_1-y_2)\ge 0.\qquad(3.5)$$

We can also suppose $\Gamma\subset\{(x,y)\,:\,|x-y|\le L\}$. We will try to "improve" the set Γ, by removing negligible sets and getting better properties. Then, we will show that the remaining set $\widetilde{\Gamma}$ is contained in the graph of a map T, thus obtaining the result.

As usual, we will assume $\mu\ll\mathscr{L}^d$. Then, we need to recall the notion of Lebesgue points, which is specific to the Lebesgue measure.

Box 3.2. – **Memo** – Density points

Definition - For a measurable set $E\subset\mathbb{R}^d$ we call Lebesgue point of E a point $x\in\mathbb{R}^d$ such that

$$\lim_{r\to 0}\frac{|E\cap B(x,r)|}{|B(x,r)|}=1.$$

The set of Lebesgue points of E is denoted by $\mathrm{Leb}(E)$ and it is well-known that $|E \setminus \mathrm{Leb}(E)| + |\mathrm{Leb}(E) \setminus E| = 0$. This is actually a consequence of a more general fact: given a function $f \in L^1_{loc}(\mathbb{R}^d)$, a.e. point x is a Lebesgue point for f, in the sense that $\lim_{r\to 0} \fint_{B(x,r)} |f(y) - f(x)|dy = 0$, which also implies $f(x) = \lim_{r\to 0} \fint_{B(x,r)} f(y)\,\mathrm{d}y$. If this is applied to $f = \mathbb{1}_E$, then one recovers the notion of Lebesgue points of a set (also called density points).

Lemma 3.22. *The plan $\overline{\gamma}$ is concentrated on a c-CM set $\widetilde{\Gamma}$ such that for every $(x_0, y_0) \in \widetilde{\Gamma}$ and for every $\varepsilon, \delta > 0$, every unit vector ξ and every sufficiently small $r > 0$, there are a point $x \in \big(B(x_0, r) \setminus B(x_0, \frac{r}{2})\big) \cap C(x_0, \xi, \delta)$ and a point $y \in B(y_0, \varepsilon)$ such that $(x, y) \in \widetilde{\Gamma}$, where $C(x_0, \xi, \delta)$ is the following convex cone:*

$$C(x_0, \xi, \delta) := \{x : (x - x_0) \cdot \xi > (1 - \delta)|x - x_0|\}.$$

Proof. First, take a c-CM set Γ such that $\overline{\gamma}(\Gamma) = 1$. Then, let B_i be a countable family of balls in $\mathbb{R}^d$, generating the topology of $\mathbb{R}^d$ (for instance, all the balls such that the coordinates of the center and the radius are rational numbers). Consider now

$$A_i := (\pi_x)(\Gamma \cap (\Omega \times B_i)),$$

i.e. the set of all points x such that at least a point y with $(x, y) \in \Gamma$ belongs to B_i (the points that have at least one "image" in B_i, if we imagine $\overline{\gamma}$ as a multi-valued map). Then set $N_i := A_i \setminus \mathrm{Leb}(A_i)$. This set has zero Lebesgue measure, and hence it is μ-negligible. Also $\mu(\bigcup_i N_i) = 0$. As a consequence, one can define $\widetilde{\Gamma} := \Gamma \setminus ((\bigcup_i N_i) \times \Omega)$. The plan $\overline{\gamma}$ is still concentrated on $\widetilde{\Gamma}$, since we only removed μ-negligible points. Obviously, $\widetilde{\Gamma}$ stays c-CM and enjoys property (3.5), since it is a subset of Γ.

Moreover, $\widetilde{\Gamma}$ has the property which is claimed in the statement. This is true since there is at least one of the balls B_i containing y_0 and contained in $B(y_0, \varepsilon)$. Since $x_0 \in A_i$ and we have removed N_i, this means that x_0 is a Lebesgue point for A_i. Since the region $\big(B(x_0, r) \setminus B(x_0, \frac{r}{2})\big) \cap C(x_0, \xi, \delta)$ is a portion of the ball $B(x_0, r)$ which takes a fixed proportion (depending on δ) of the volume of the whole ball, for $r \to 0$ it is clear that A_i (and also $\mathrm{Leb}(A_i)$) must meet it (otherwise x_0 would not be a Lebesgue point). It is then sufficient to pick a point in $\mathrm{Leb}(A_i) \cap \big(B(x_0, r) \setminus B(x_0, \frac{r}{2})\big) \cap C(x_0, \xi, \delta)$ and we are done. □ □

Lemma 3.23. *If (x_0, y_0) and (x_0, z_0) belong to $\widetilde{\Gamma}$, then $y_0 = z_0$.*

Proof. Suppose by contradiction $y_0 \neq z_0$. In order to fix the ideas, let us suppose $|x_0 - z_0| \leq |x_0 - y_0|$ (and in particular $y_0 \neq x_0$, since otherwise $|x_0 - z_0| = |x_0 - y_0| = 0$ and $z_0 = y_0$).

Now, use the property of $\widetilde{\Gamma}$ and find $(x, y) \in \widetilde{\Gamma}$ with $y \in B(y_0, \varepsilon)$ and $x \in \big(B(x_0, r) \setminus B(x_0, \frac{r}{2})\big) \cap C(x_0, \xi, \delta)$, for a vector ξ to be determined later. Use now the fact that $\widetilde{\Gamma}$ is c-CM applied to (x_0, z_0) and (x, y).

If we can prove that $|x-z_0|, |x_0-y| \leq L$, then we should have

$$(x-x_0)\cdot(y-z_0) \geq 0$$

and we would use this to get a contradiction, by means of a suitable choice for ξ. Indeed, the direction of $x-x_0$ is almost that of ξ (up to an error of δ) and that of $y-z_0$ is almost that of y_0-z_0 (up to an error of the order of $\varepsilon/|z_0-y_0|$). If we choose ξ such that $\xi\cdot(y_0-z_0) < 0$, this means that for small δ and ε we would get a contradiction.

We need to guarantee $|x-z_0|, |x_0-y| \leq L$, in order to prove the claim. Let us compute $|x-z_0|^2$: we have

$$|x-z_0|^2 = |x_0-z_0|^2 + |x-x_0|^2 + 2(x-x_0)\cdot(x_0-z_0).$$

In this sum we have

$$|x_0-z_0|^2 \leq L^2; \quad |x-x_0|^2 \leq r^2; \quad 2(x-x_0)\cdot(x_0-z_0) = |x-x_0|(\xi\cdot(x_0-z_0)+O(\delta)).$$

Suppose now that we are in one of the following situations: either choose ξ such that $\xi\cdot(x_0-z_0) < 0$ or $|x_0-z_0|^2 < L^2$. In both cases we get $|x-z_0| \leq L$ for r and δ small enough. In the first case, since $|x-x_0| \geq \frac{r}{2}$, we have a negative term of the order of r and a positive one of the order of r^2; in the second we add to $|x_0-z_0|^2 < L^2$ some terms of the order of r or r^2.

Analogously, for what concerns $|x_0-y|$ we have

$$|x_0-y|^2 = |x_0-x|^2 + |x-y|^2 + 2(x_0-x)\cdot(x-y).$$

The three terms satisfy

$$|x_0-x|^2 \leq r^2; \quad |x-y|^2 \leq L^2;$$
$$2(x_0-x)\cdot(x-y) = |x-x_0|(-\xi\cdot(x_0-y_0)+O(\delta+\varepsilon+r)).$$

In this case, either we have $|x_0-y_0| < L$ (which would guarantee $|x_0-y| < L$ for ε, δ small enough), or we need to impose $\xi\cdot(y_0-x_0) < 0$.

Note that imposing $\xi\cdot(y_0-x_0) < 0$ and $\xi\cdot(x_0-z_0) < 0$ automatically gives $\xi\cdot(y_0-z_0) < 0$, which is the desired condition so as to have a contradiction. Set $\mathbf{v} = y_0-x_0$ and $\mathbf{w} = x_0-z_0$. If it is possible to find ξ with $\xi\cdot\mathbf{v} < 0$ and $\xi\cdot\mathbf{w} < 0$ we are done. When is it the case that two vectors $\mathbf{v}$ and $\mathbf{w}$ do not admit the existence of a vector ξ with both scalar products that are negative? The only case is when they go in opposite directions. But this would mean that x_0, y_0 and z_0 are colinear, with z_0 between x_0 and y_0 (since we supposed $|x_0-z_0| \leq |x_0-y_0|$). If we want z_0 and y_0 to be distinct points, we should have $|x_0-z_0| < L$. Hence in this case we do not need to check $\xi\cdot(x_0-z_0) < 0$. We only need a vector ξ satisfying $\xi\cdot(y_0-x_0) < 0$ and $\xi\cdot(y_0-z_0) < 0$, but the directions of y_0-x_0 and y_0-z_0 are the same, so that this can be guaranteed by many choices of ξ, since we only need the scalar product with one only direction to be negative. Take for instance $\xi = -(y_0-x_0)/|y_0-x_0|$. □ □

Theorem 3.24. *The secondary variational problem*

$$\min\{K_2(\gamma) : \gamma \in O_\infty(\mu,\nu)\}$$

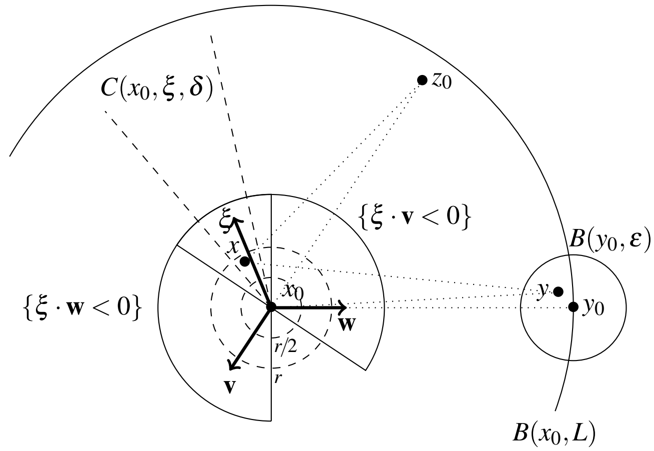

Figure 3.4: The points in the proof of Lemma 3.23.

admits a unique solution $\overline{\gamma}$, it is induced by a transport map T*, and such a map is an optimal transport for the problem*

$$\min\{||\mathrm{T}-\mathrm{id}||_{L^\infty(\mu)}\ ,\ \mathrm{T}_\#\mu=\nu\}.$$

Proof. We have already seen that $\overline{\gamma}$ is concentrated on a set $\widetilde{\Gamma}$ satisfying some useful properties. Lemma 3.23 shows that $\widetilde{\Gamma}$ is contained in a graph, since for any x_0 there is no more than one possible point y_0 such that $(x_0,y_0)\in\widetilde{\Gamma}$. Let us consider such a point y_0 as the image of x_0 and call it $\mathrm{T}(x_0)$. Then $\overline{\gamma}=\gamma_\mathrm{T}$. The optimality of T in the "Monge" version of this L^∞ problem comes from the usual comparison with the Kantorovich version on plans γ. The uniqueness comes from standard arguments (since it is true as well that convex combinations of minimizers should be minimizers, and this allows to perform the proof in Remark 1.19). □ □

3.3 Discussion

3.3.1 Different norms and more general convex costs

The attentive reader will have observed that the proof in Section 3.1 about the case of the distance cost function was specific to the case of the distance induced by the Euclidean norm. Starting from the original problem by Monge, and relying on the proof strategy by Sudakov [290] (which had a gap later solved by Ambrosio in [8, 11], but was meant to treat the case of an arbitrary norm), the case of different norms has been extensively studied in the last years. Note that the case of uniformly convex norms (i.e. those such that the Hessian of the square of the norm is bounded from below by a matrix which is positively definite) is the one which is most similar to the Euclidean case, and can be treated in a similar way.